Sri Chaitanya IIT Academy., India.

AP, TELANGANA, KARNATAKA, TAMILNADU, MAHARASHTRA, DELHI, RANCHI

A right Choice for the Real Aspirant

ICON Central Office, Madhapur – Hyderabad

Sec: Sr-ICON-B-1 | **JEE-ADVANCE-2019-P2** | **Date: 02-05-21**

Time: 02:30 PM to 05:30 PM | **GTA-02** | **Max.Marks:186**

SYLLABUS:

Physics : TOTAL SYLLABUS

Chemistry : TOTAL SYLLABUS

Mathematics : TOTAL SYLLABUS

Name of the Student: ____________________ **H.T. NO:** | | | | | | | |

JEE-ADVANCE-2019-P2-Model

Time: 3:00Hr's **IMPORTANT INSTRUCTIONS** Max Marks: 186

PHYSICS:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N : 1 – 8)	Questions with Multiple Correct Choice **+1 partial marks**	4	-1	8	32
Sec – II(Q.N : 9 – 14)	Questions with Numerical Value Answer Type	3	0	6	18
Sec – III(Q.N : 15 – 18)	Matching Type	3	-1	4	12
Total				**18**	**62**

CHEMISTRY:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N : 1 – 8)	Questions with Multiple Correct Choice **+1 partial marks**	4	-1	8	32
Sec – II(Q.N : 9 – 14)	Questions with Numerical Value Answer Type	3	0	6	18
Sec – III(Q.N : 15 – 18)	Matching Type	3	-1	4	12
Total				**18**	**62**

MATHEMATICS:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N : 1 – 8)	Questions with Multiple Correct Choice **+1 partial marks**	4	-1	8	32
Sec – II(Q.N : 9 – 14)	Questions with Numerical Value Answer Type	3	0	6	18
Sec – III(Q.N : 15 – 18)	Matching Type	3	-1	4	12
Total				**18**	**62**

PHYSICS **Max Marks: 62**

SECTION – I

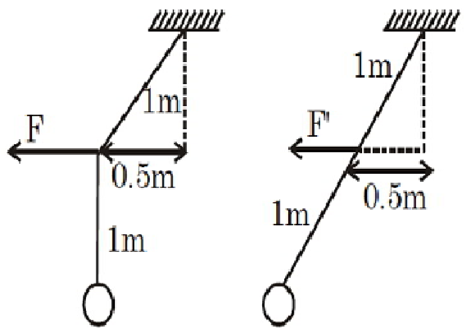
(ONE OR MORE CORRECT ANSWER TYPE)

This section contains 8 multiple choice questions. Each question has 4 options (A), (B), (C) and (D) for its answer, out of which ONE OR MORE than ONE option can be correct. **Marking scheme: +4 for all correct options & +1 partial marks, 0 if not attempted and -1 in all wrong cases**

Section 1 (Max Marks: 32)

- Section 1 contains eight questions.
- Each Question has Four Options and Only One of these four will be the correct answer.
- For each question, choose the option corresponding to the correct answer
- The Marking scheme to evaluate the Answer to each question:
  Full Marks: +4 (If the answer is correct)
  Partial Marks: +3 (If all the four options are correct but only 3 are chosen)
  Partial Marks: +2 (If THREE or More options are correct but only 2 correct answers are chosen)
  Partial Marks: +1 (If TWO or More options are correct but only 1 correct answer is chosen)
  Zero Marks: 0 (If the question is unanswered)
  Negative Marks: -1 (In all other cases)

1 A bob of mass 1 kg is hanging from an inextensible massless string of length 2m. The string is pulled slowly at the midpoint to the left so that both portions of string are taut. The midpoint is now held at 0.5 m away from the original vertical position by a horizontal force. The same experiment is repeated, but with a hinged massless rod replacing the string:

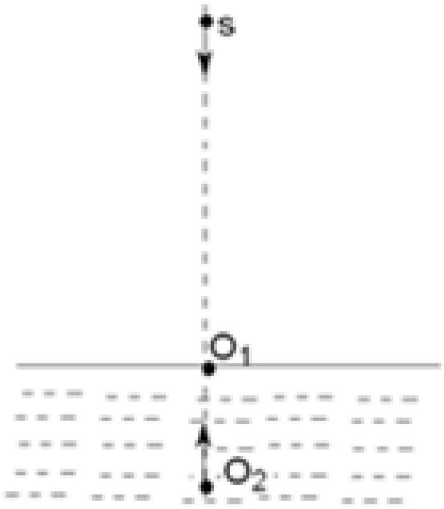


A. The force required to hold the midpoint is same in both cases.
B. The force required to hold the midpoint is larger in second case.
C. The tension in the string between the midpoint and the hinge is less than the tension in the rod between the midpoint and the hinge
D. The tension in the string between the midpoint and the bob is more than the tension in the rod between the midpoint and the bob.

2 Two spherical black-bodies A and B, having radii r_A and r_B, where $r_B = 2r_A$ emit radiations with peak intensities at wavelengths 400 nm and 800 nm respectively. If their temperature are T_A and T_B respectively in Kelvin scale, their emissive powers are E_A and E_B and energies emitted per second are P_A and P_B then :

A. $T_A / T_B = 2$ **B.** $P_A / P_B = 4$ **C.** $E_A / E_B = 8$ **D.** $E_A / E_B = 4$

3 In the figure shown an observer O_1 floats (at rest) on water surface while another observer O_2 is moving upwards with constant velocity $v_1 = \frac{v}{5}$ in water. The source (s) moves down with constant velocity $v_s = \frac{v}{5}$ and emits sound of frequency f. The velocity of sound in air is v and that in water is 4v. Then,



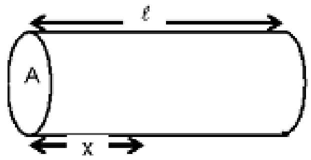
A. the wave length of the sound received by O_1 is $\frac{4v}{5f}$

B. the wavelength of the sound received by O_1 is $\frac{v}{f}$

C. the frequency of the sound received by O_2 is $\frac{21f}{16}$

D. the wavelength of the sound received by O_2 is $\frac{4v}{7f}$

4 The cross-sectional area and length of cylindrical conductor are A and ℓ respectively. The specific conductivity varies as $\sigma(x) = \sigma_0 \ell / \sqrt{x}$ where x is the distance along the axis of the cylinder from one of its ends. A potential difference V_0 is applied across the cylinder. Choose the correct statement(s)



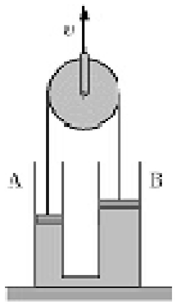
A. The resistance of the system along the cylindrical axis is $\frac{2\sqrt{\ell}}{3A\sigma_0}$

B. The current density is constant throughout the length of the cylinder.

C. The electric field E(x) at each point in the cylinder is $E(x)=\frac{5V_0\sqrt{x}}{2\ell^{3/2}}$

D. The electric field E(x) at each point in the cylinder is $E(x)=\frac{3V_0\sqrt{x}}{2\ell^{3/2}}$

5 Two vertical cylindrical vessels A and B of horizontal cross-sectional areas S and 2S are connected at their bottoms with a horizontal tube of cross-sectional area 0.5S. An amount of water is trapped in the vessels under leak proof pistons, one in each cylindrical vessel. The pistons are connected with a light inextensible thread that passes over an ideal pulley as shown in the figure. The pulley is pulled upwards with a constant velocity υ. The vessels are rigidly affixed on the horizontal floor. Mark the correct statements.



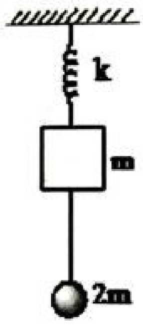
A. Piston A will shift upwards and the piston B downwards.
B. Speeds of the pistons A and B are 4υ and 2υ respectively
C. Flow velocity in the horizontal connecting tube is rightwards.
D. Flow velocity in the horizontal connecting tube is 8υ.

6 Nucleus A decays to B with decay constant λ_1 and B decays to C with a decay constant λ_2. Initially at t = 0 number of nuclei of A and B are $2N_0$ and N_0 respectively. At $t=t_0$, number of nuclei B stop changing. If at this instant number of nuclei of B are $\frac{3N_0}{2}$, then

A. $t_0 = \frac{1}{\lambda_1}\ln\frac{4\lambda_1}{3\lambda_2}$ **B.** $t_0 = \frac{1}{\lambda_2}\ln\frac{4\lambda_1}{3\lambda_2}$

C. $N_A = \frac{3N_0}{2}\frac{\lambda_2}{\lambda_1}$ at $t = t_0$ **D.** $N_A = \frac{2N_0}{3}\frac{\lambda_2}{\lambda_1}$ at $t = t_0$

7 A bob of mass 2m hangs by a string attached to the block of mass m of a spring blocks system. The whole arrangement is in a state of equilibrium. The bob of mass 2m is pulled down slowly by a distance x_0 and released



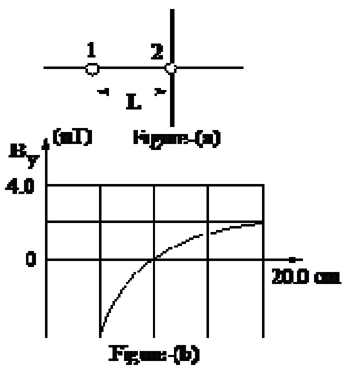
A. For $x_0 = \frac{3mg}{k}$, maximum tension in string is 4 mg

B. For $x_0 > \frac{3mg}{k}$, minimum tension in string is mg

C. Frequency of oscillation of system is $\frac{1}{2\pi}\sqrt{\frac{k}{3m}}$, for all non-zero values of x_0

D. The motion will remain simple harmonic for $x_0 \le \frac{3mg}{k}$.

8 In the adjoining figure (a) two long parallel wires carrying current and separated by a distance L. The ratio of current I_1 and I_2 is given as 4.00. The direction of current are not indicated. In the figure (b) the y-component of B_y of their net magnetic field along x-axis to the right of wire 2.



A. The maximum value of B_y is at x = 30.0 cm to the right of wire 2

B. If I_2 is 3mA, the maximum value of B_y is 2.0 nT

C. Direction of current in wire 1 is out of the page

D. The measurement of L is 30.0 cm

SECTION – II
(NUMERICAL VALUE TYPE)

This section contains 6 questions. Each question is numerical value type. For each question, enter the correct numerical value
(e.g. 003.2343, 6.250023, 7.000000, -0.33, -.30, 30.27000, -127.0030).
Marking scheme: +3 for correct answer, 0 if not attempted and 0 in all other cases. Section 2 (Max Marks: 18)

- Section two contains six (6) questions. The answer to each question is a NUMERICAL VALUE
- For each question enter the correct numerical value of the answer using the mouse and the on-screen virtual numerical keypad in the place designated to enter the answer. If the numerical values has more than two decimal places, truncate/round off the value to TWO decimal places.

Answer to each question will be evaluated according to the following marking scheme
Full Marks: +3 (If only the correct numerical value is entered)
Zero Marks: 0 (In all other cases)

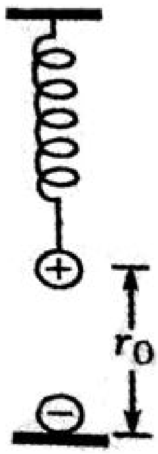
9 A rod of length L is pivoted at one end and its rotated with a uniform angular velocity in horizontal lane. Let T_1 and T_2 be the tensions at the point L/4 and 3L/4 away from the pivoted ends. Then $\frac{T_1}{T_2}$ is nearly

10 In an experiment to determine the value of acceleration due to gravity g using a simple pendulum, the measured value of length of the pendulum is 31.4cm known to 1 mm accuracy and the time period for 100 oscillations of pendulum is 112.0s known to 0.01 s accuracy. Find the accuracy in determining the value of g.

11 In a nuclear reactor an element X decays to a radio active element Y at a constant rate 10^{15} atoms per sec. Each decay releases 100 MeV energy. Half life of Y equals T and decays to a stable product Z. Each decay of Y releases 50 MeV. All energy released inside the reactor is used to produce electricity at an efficiency of 30%. Calculate the electrical power in kW generated in the reactor in steady state.

12 The magnetic field in region is given by $\bar{B} = B_0\left(\frac{x}{a}\right)\hat{k}.A$ square loop of side d is placed with its edges along x-axis and y-axis. The loop is moved with a constant velocity $\bar{V} = V_0\hat{i}$. The emf induced in the loop is $\frac{B_0V_0^x d^y}{a^z}$,then $\frac{xy}{z}$ is equal to

13 A positive charged sphere of mass $m = 5kg$ is attached by a spring of force constant $K = 10^4\,Nm^{-1}$. The sphere is tied with a thread so that spring is in its natural length. Another identical, negatively charged sphere is fixed with floor, vertically below the positively charged sphere as shown is figure. If initial separation between spheres is $r_0 = 50cm$ and magnitude of charge on each sphere is $q = 100\mu C$. The maximum elongation of spring when the thread is burnt is $P \times 5cm$. Find $P\left(g = 10ms^{-2}\right)$



14 Two guns situated on the top of a hill of height 10m fire one shot each with the same speed $5\sqrt{3}$ m/s at some interval of time. One gun fires horizontally and other fires upwards at an angle of 60^0 with the horizontal. The shots collide at point P. The coordinates of point P with respect to the bottom of the hill are (in metres x,y). find $\sqrt{3}x + y$

SECTION-III
(MATCHING LIST TYPE)

This section contains 4 questions, each having two matching lists (List-1 & List-II). The options for the correct match are provided as (A), (B),(C) and (D) out of which ONLY ONE is correct.
Marking scheme: +3 for correct answer, 0 if not attempted and -1 in all other cases.
Section 3 (Max Marks: 12)

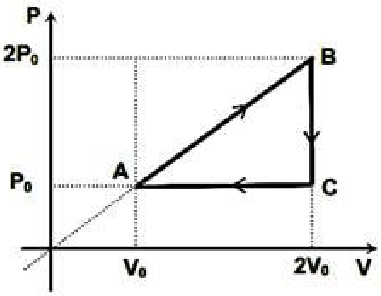
- This Section contains two (02) List-Match sets.
- Each List-Match set has TWO (02) Multiple-choice Questions
- Each List-Match set has two lists: List 1 and List 2
- List 1 has four entries (I), (II), (III), (IV) and List 2 has six entries (P), (Q), (R), (S), (T) and (U)
- Four options are given in each multiple choice question based on List 1 and List 2 and Only ONE of these four options satisfies the condition asked in the Multiple Choice Question.

Answer to each question will be evaluated according to the following marking scheme
Full Marks: +3 (If only the option corresponding to the correct combination is chosen))
Zero Marks: 0 (If none of the options is chosen (i.e. the question is unanswered)
Negative Marks: -1 (In all other cases)

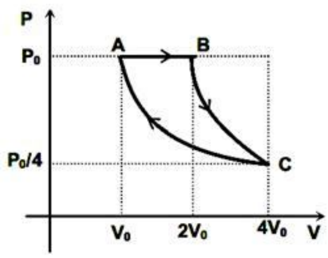
Answer Q.15 and Q.16 by appropriately matching the list based on the information given in the paragraph

One mole of an ideal gas is taken through four different cyclic processes as depicted in the **List-I** and **List-II** gives the values of total work done and total heat absorbed by the gas during four different cyclic processes

List-I **List-II**

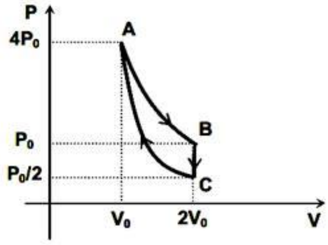
(I) 

monoatomic gas is taken (P) $\frac{P_0V_0}{2}$

(II) 

monoatomic gas is taken

(For the process BC, PV^2 = constant and process CA is an isothermal process) (Q) P_0V_0

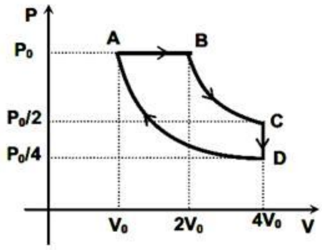


(III)

Diatomic gas is taken

(For the process AB, PV^2 = constant

and for the process CA, PV^3 = constant) **(R)** $2P_0V_0(1-\ell n2)$



(IV)

Diatomic gas is taken

(Processes BC and DA are isothermal) **(S)** $\frac{5P_0V_0}{2}$

(T) $6P_0V_0$

(U) $\frac{P_0V_0}{2}(7+4\ell n2)$

15 Find the **CORRECT** match for the total work done by the gas during each cyclic process

A. (I) → (S), (II) → (R), (III) → (P), (IV) → (T)

B. (I) → (P), (II) → (R), (III) → (P), (IV) → (Q)

C. (I) → (P), (II) → (Q), (III) → (R), (IV) → (T)

D. (I) → (Q), (II) → (P), (III) → (S), (IV) → (R)

16 Find the **CORRECT** match for the total heat absorbed by the gas during each cyclic process

A. (I) → (Q), (II) → (S), (III) → (T), (IV) → (P)

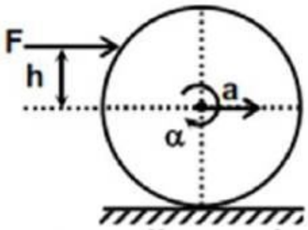
B. (I) → (T), (II) → (S), (III) → (Q), (IV) → (R)

C. (I) (EQTN) (T), (II) → (S), (III) → (T), (IV) → (U)

D. (I) → (P), (II) → (S), (III) → (R), (IV) → (Q)

Answer Q.17 and Q.18 by appropriately matching the list based on the information given in the paragraph

17 A solid body of mass m and radius 'R' is placed on a rough horizontal surface. When a horizontal force 'F' is applied to the body at a height 'h' above the centre level, the body starts rolling without slipping on the horizontal surface as shown.



List-I gives the four different shapes of the given solid body and corresponding values of F and h while List-II gives the magnitude of some quantities.

List – I	List – II
(I)For a ring, $F = 2mg, h = \frac{R}{2}$	(P) 1
(II)For a disc, $F = mg, h = \frac{3R}{4}$	(Q) $\frac{1}{3}$
(III)For a hollow sphere, $F = \frac{5mg}{8}, h = \frac{R}{3}$	(R) $\frac{7}{9}$
(IV)For a solid sphere, $F = \frac{mg}{2}, h = \frac{3R}{5}$	(S) $\frac{8}{21}$
	(T) $\frac{1}{4}$
	(U) $\frac{1}{7}$

17 If acceleration of the ring is $'a_0'$ then the acceleration of the different shapes of the body in a_0 units will be correctly matched in

A. $I \to P, II \to R, III \to Q, IV \to S$ **B.** $I \to Q, II \to P, III \to S, IV \to U$

C. $I \to P, II \to T, III \to R, IV \to S$ **D.** $I \to T, II \to Q, III \to R, IV \to U$

18 If magnitude of frictional force acting on the ring is $'f_0'$ then the magnitude of frictional force acting on the different shapes of the body in $'f_0'$ units will be frictional force acting on the different shapes of the body in $'f_0'$ units will be

A. $I \to P, II \to Q, III \to R, IV \to S$ **B.** $I \to Q, II \to P, III \to S, IV \to U$

C. $I \to P, II \to Q, III \to T, IV \to U$ **D.** $I \to T, II \to Q, III \to R, IV \to U$

CHEMISTRY

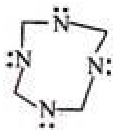
SECTION – I
(ONE OR MORE CORRECT ANSWER TYPE)

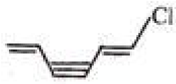
This section contains 8 multiple choice questions. Each question has 4 options (A), (B), (C) and (D) for its answer, out of which ONE OR MORE than ONE option can be correct. **Marking scheme: +4 for all correct options & +1 partial marks, 0 if not attempted and -1 in all wrong cases**

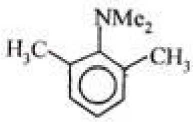
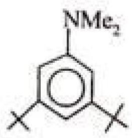
Section 1 (Max Marks: 32)

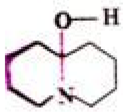
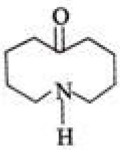
- Section 1 contains eight questions.
- Each Question has Four Options and Only One of these four will be the correct answer.
- For each question, choose the option corresponding to the correct answer
- The Marking scheme to evaluate the Answer to each question:
  Full Marks: +4 (If the answer is correct)
  Partial Marks: +3 (If all the four options are correct but only 3 are chosen)
  Partial Marks: +2 (If THREE or More options are correct but only 2 correct answers are chosen)
  Partial Marks: +1 (If TWO or More options are correct but only 1 correct answer is chosen)
  Zero Marks: 0 (If the question is unanswered)
  Negative Marks: -1 (In all other cases)

19 Which of the following statements are incorrect?

A.  is more basic than $(CH_3)_3N$:

B.  forms white precipitate with $Ag^+(aq)$ most readily.

C.  is more basic than 

D.  and  re resonating structure to each other.

20 Which of the following are ambidient nucleophiles?

A. OH^- **B.** NO_2^- **C.** N_2H_4 **D.** NCO^-

21 $CH_3-CH_2-CH_2-\overset{\overset{O}{\|}}{C}-CH_3 \xrightarrow{Conc\ HNO_3}$

Products , what are the products

A. propanoic acid **B.** Ethanoic acid **C.** Butanoic acid **D.** Methanoic acid

22 $C(s)+H_2O(g) \rightarrow CO(g)+H_2(g)$;

$\Delta H^\circ = +131\ kJ;\ \Delta S^\circ = +134\ J/K$

Mark out the correct statement(s)

A. Reaction is spontaneous even at the room temperature.

B. Reaction is not spontaneous at the room temperature.

C. Reaction is spontaneous above $705^\circ C$.

D. ΔH outweighs the entropy factor $T\Delta S$ at the room temperature.

23 20 ml of 0.2M $Al_2(SO_4)_3$ is mixed with 20 ml of 0.6M $BaCl_2$. The resultant solution is

A. 0. 9 M **B.** 0.2 M $Al^{3+}\ ions$ **C.** 0.6 M Cl^- ions **D.** 0.12 N

24 In correct statements

A. In a closed system G is always minimum at equilibrium

B. A graph plotted between log Kc and $\frac{1}{T}$ is straight line with intercept 10 and slope equal to 0.5. Then heat of reaction 2.303

C. The Kc for the reaction between I_2 and Cs_2 and H_2O is 588 in favour of Cs_2, one lit of aquous solution containing ore gm of I_2 is shaken with 50ml. Cs_2, the amount of I_2 left is 0.035 g

D. In a systems $A(s) \rightleftarrows 2B(g)+3c(g)$ Concentration of 'C' at equilibrium is increased by a factor '2' it will cause equilibrium concentration of 'B' to change to $2\sqrt{2}$ times its original value

25 Identify the CORRECT statements among the following

A. Cr^{2+} is a stronger reducing agent than Fe^{2+} in aqueous solution

B. E^0 for the redox couple $Mn^{3+}|Mn^{2+}$ is much more positive than that for $Fe^{3+}|Fe$

C. A metal exhibits its lowest oxidation state in its binary compound with fluorine

D. Zinc has lowest enthalpy of atomization among 3d-series elements

26 $Na_2[B_4O_5(OH)_4]\cdot 8H_2O$ called borax. Select correct for borax.

A. On heating glassy solid is obtained which is a composition of $NaBO_2$ and B_2O_3.

B. All borons use sp^3 orbital's for bonding.

C. Its aqueous solution is alkaline in nature.

D. Its aqueous solution produced boric acid when treated with conc. H_2SO_4

SECTION – II
(NUMERICAL VALUE TYPE)

This section contains 6 questions. Each question is numerical value type. For each question, enter the correct numerical value
(e.g. 003.2343, 6.250023, 7.000000, -0.33, -.30, 30.27000, -127.0030).
Marking scheme: +3 for correct answer, 0 if not attempted and 0 in all other cases. Section 2 (Max Marks: 18)

- Section two contains six (6) questions. The answer to each question is a NUMERICAL VALUE
- For each question enter the correct numerical value of the answer using the mouse and the on-screen virtual numerical keypad in the place designated to enter the answer. If the numerical values has more than two decimal places, truncate/round off the value to TWO decimal places.

Answer to each question will be evaluated according to the following marking scheme
Full Marks: +3 (If only the correct numerical value is entered)
Zero Marks: 0 (In all other cases)

27 If equivalent conductance of 1 M benzoic acid is $12.8\,ohm^{-1}cm^{2}$ and if the conductance of benzoate ion and H^{+} ion are 42 and $288.42\ ohm^{-1}cm^{2}$ respectively, its degree of dissociation is around___%

28 1.60g of a metal was dissolved in HNO_3 to prepare its nitrate. The nitrate on strong heating gives 2g oxide. The equivalent weight of metal is 4x, then x is

29 One gram of charcoal adsorbs $100ml$ 0.5M CH_3COOH and then molarity of CH_2COOH reduces to 0.49 M. The number of milli moles of Acetic acid adsorbs is ___

30 0.002 gm of an organic compound was treated according to Kjeldhal's method. 0.2×10^{-4} mol of H_2SO_4 was required to neutralise. NH_3. The percentage of N_2 is 7x %, then x is

31 One mole of 1,2-Dibromopropane on treatment with x moles of $NaNH_2$ followed by treatment with ethyl bromide gave 2- pentyne. The value of is x

32 Cl_2 gas when reacted with excess of cold $NaOH$ fraction of chlorine oxidised is 'a' moles while Cl_2 gas when reacted with excess of hot NaOH fraction of chlorine oxidised is 'b' moles . The value of $\frac{a}{b}$ is ________

SECTION-III
(MATCHING LIST TYPE)

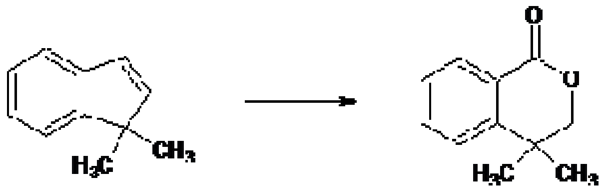
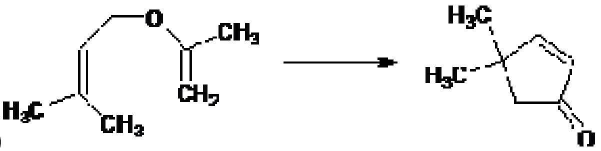
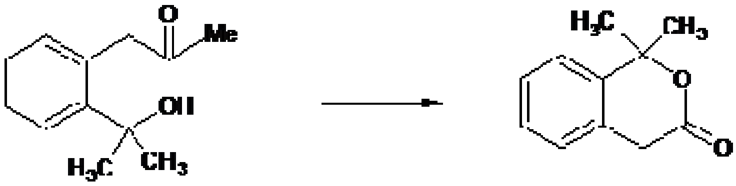
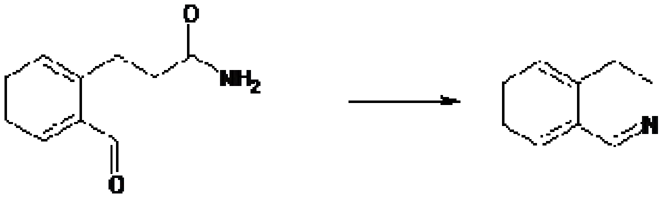
This section contains 4 questions, each having two matching lists (List-1 & List-II). The options for the correct match are provided as (A), (B),(C) and (D) out of which ONLY ONE is correct.
Marking scheme: +3 for correct answer, 0 if not attempted and -1 in all other cases.
Section 3 (Max Marks: 12)

- This Section contains two (02) List-Match sets.
- Each List-Match set has TWO (02) Multiple-choice Questions
- Each List-Match set has two lists: List 1 and List 2
- List 1 has four entries (I), (II), (III), (IV) and List 2 has six entries (P), (Q), (R), (S), (T) and (U)
- Four options are given in each multiple choice question based on List 1 and List 2 and Only ONE of these four options satisfies the condition asked in the Multiple Choice Question.

Answer to each question will be evaluated according to the following marking scheme
Full Marks: +3 (If only the option corresponding to the correct combination is chosen))
Zero Marks: 0 (If none of the options is chosen (i.e. the question is unanswered)
Negative Marks: -1 (In all other cases)

Answer Q.33 and Q.34 by appropriately matching the list based on the information given in the paragraph

List – I includes starting materials and products of selected chemical reactions. List-II gives reagent that may be used to carry out reactions of List –I.

List – I	**List – II**
(I) 	(P) $NaOH + Br_2 / \Delta$
(II) 	(Q) $conc\, H_2SO_4 / \Delta$
(III) 	(R) $NaIO_4$
(IV) 	(S) $dil.cold\ KMnO_4$
	(T) $NaOH / \Delta$
	(U)Heat

33 Which of the following options has correct combination of Reagents required in correct sequence to carry out List I reactions?

A. I- RTQ **B.** III- TSU **C.** II-USRT **D.** IV- TRU

34 Which of the following options has correct combination of Reagents required in correct sequence to carry out List I reactions?

A. III-QSU **B.** I-SRTQ **C.** IV-PS **D.** II-QTU

Answer Q.35 and Q.36 by appropriately matching the list based on the information given in the paragraph

Consider the Bohr's model of a one-electron system where the electron moves around the nucleus of charge Z. In the following, List-I contains some quantities for the n^{th}, orbit of the atom and List-II contains options showing how they depend on *n and Z*

List-I	List-II
(I)Angular momentum of the electron in the n^{th} orbit	(P) $\propto nZ^0$
(II)de-Broglie wavelength of the electron in the n^{th} orbit	(Q) $\propto nZ^{-1}$
(III)Energy required to remove an electron from the n^{th} orbit	(R) $\propto n^2Z^{-2}$
(IV)The number of revolutions of an electron in the n^{th} orbit per second	(S) $\propto n^{-2}Z^2$
(V)Velocity of an electron in the n^{th} orbit	(T) $\propto n^{-3}Z^2$
	(U) $\propto n^{-1}Z$

35 Which of the following options has the correct combination considering List I and List II?

A. (I) (Q) **B.** (II) (P) **C.** (III) (R) **D.** (IV) (T)

36 Which of the following options has the correct combination considering List I and List II?

A. (I) (Q) **B.** (II) (P) **C.** (III) (R) **D.** (IV) (T)

MATHEMATICS

SECTION – I

(ONE OR MORE CORRECT ANSWER TYPE)

This section contains 8 multiple choice questions. Each question has 4 options (A), (B), (C) and (D) for its answer, out of which ONE OR MORE than ONE option can be correct. **Marking scheme: +4 for all correct options & +1 partial marks, 0 if not attempted and -1 in all wrong cases**

Section 1 (Max Marks: 32)

- Section 1 contains eight questions.
- Each Question has Four Options and Only One of these four will be the correct answer.
- For each question, choose the option corresponding to the correct answer
- The Marking scheme to evaluate the Answer to each question:
  Full Marks: +4 (If the answer is correct)
  Partial Marks: +3 (If all the four options are correct but only 3 are chosen)
  Partial Marks: +2 (If THREE or More options are correct but only 2 correct answers are chosen)
  Partial Marks: +1 (If TWO or More options are correct but only 1 correct answer is chosen)
  Zero Marks: 0 (If the question is unanswered)
  Negative Marks: -1 (In all other cases)

37 If a vector $\bar{r}$ satisfies the equation $\bar{r}\times\left(\hat{i}+2\hat{j}+\hat{k}\right)=\hat{i}-\hat{k}$, then $\bar{r}$ is equal to

A. $\hat{i}+3\hat{j}+\hat{k}$

B. $3\hat{i}+7\hat{j}+3\hat{k}$

C. $\hat{j}+t\left(\hat{i}+2\hat{j}+\hat{k}\right)$ where t is any scalar

D. $\hat{i}+(t+3)\hat{j}+\hat{k}$ where t is any scalar

38 If A,B are non singular matrices of same order , then $Adj\,AB=$

A. $(AdjB).(AdjA)$ **B.** $|B||A|B^{-1}A^{-1}$ **C.** $|B||A|A^{-1}B^{-1}$ **D.** $|A||B|(AB)^{-1}$

39 If $\sin\beta$ is the geometric mean between $\sin\alpha$ *and* $\cos\alpha$, then $\cos 2\beta$ is equal to

A. $2\sin^2\left(\frac{\pi}{4}-\alpha\right)$ **B.** $2\cos^2\left(\frac{\pi}{4}-\alpha\right)$ **C.** $2\cos^2\left(\frac{\pi}{4}+\alpha\right)$ **D.** $2\sin^2\left(\frac{\pi}{4}+\alpha\right)$

40 If $\lim\limits_{x\to 0}\frac{x(1+a\cos x)-b\sin x}{x^3}=1$, then

A. $b-a=5/6$ **B.** $a=-5/2$ **C.** $a-b=-1$ **D.** $b=3/2$

41 If a, b, c are natural numbers and $\frac{ax^4-bx^3+cx^2-bx+a}{\left(x^2+1\right)^2}$ attains minimum value at $x=2$ or $x=\frac{1}{2}$, then the possible values of a, b, c are respectively

A. $1,4,7$ **B.** $1,8,12$ **C.** $2,4,9$ **D.** $1,2,3$

42 Let $f(x)=\cos x\sin 2x$, then

A. $\min\limits_{x\in(-\pi,\pi)} f(x) > -7/9$

B. $\min\limits_{x\in(-\pi,\pi)} f(x) > -9/7$

C. $\min\limits_{x\in[-\pi,\pi]} f(x) > -1/9$

D. $\min\limits_{x\in[-\pi,\pi]} f(x) > -2/9$

43 The system of equations

$(a\alpha+b)x+ay+bz=0$

$(b\alpha+c)x+by+cz=0$

$(a\alpha+b)y+(b\alpha+c)z=0$

Has a non-trivial solution, if

A. a,b,c *are in A.P.*

B. a,b,c *are in G.P.*

C. a,b,c *are in H.P.*

D. α *is a root of* $ax^2+2bx+c=0$

44 $f(x)=\min\{1,\cos x,1-\sin x\}, -\pi\le x\le\pi$ then

A. $f(x)$ is not differentiable at 0

B. $f(x)$ is differentiable at $\frac{\pi}{2}$

C. $f(x)$ has local maxima at 0

D. None of these

SECTION – II
(NUMERICAL VALUE TYPE)

This section contains 6 questions. Each question is numerical value type. For each question, enter the correct numerical value

(e.g. 003.2343, 6.250023, 7.000000, -0.33, -.30, 30.27000, -127.0030).

Marking scheme: +3 for correct answer, 0 if not attempted and 0 in all other cases. Section 2 (Max Marks: 18)

- Section two contains six (6) questions. The answer to each question is a NUMERICAL VALUE
- For each question enter the correct numerical value of the answer using the mouse and the on-screen virtual numerical keypad in the place designated to enter the answer. If the numerical values has more than two decimal places, truncate/round off the value to TWO decimal places.

Answer to each question will be evaluated according to the following marking scheme

Full Marks: +3 (If only the correct numerical value is entered)

Zero Marks: 0 (In all other cases)

45 $\int_{-a}^{a}\frac{x^2e^{x^3}-e^{-x^3}}{e^{x^3}+e^{-x^3}}dx=$ ____________

46 Let $F=2i+2j+5k$ and $\bar{A}=(1,2,5)$, $\bar{B}=(-1,-2,-3)$ and

$\overrightarrow{BA}\times F=4i+6j+2\lambda k$, then the value of $|\lambda|$ is

47 Find the number of integral solutions of $x+y+z\le 29$ such that $x>0$, $y>1$, $z>2$.

48 If $l_i^2+m_i^2+n_i^2=1(i=1,2,3)$, and $l_1l_2+m_1m_2+n_1n_2=0$, $l_2l_3+m_2m_3+n_2n_3=0$, $l_3l_1+m_1m_3+n_1n_3=0$ and $\Delta=\begin{vmatrix} l_1 & m_1 & n_1 \\ l_2 & m_2 & n_2 \\ l_3 & m_3 & n_3 \end{vmatrix}$ then absolute value of Δ is ___.

49 Three six-faced fair die are thrown together let $P(k)$ denote the probability that the sum of the numbers appearing on the dice is $k(9 \le k \le 14)$. Find 54S . where $S=\sum_{k=9}^{14} P(k)$.

50 If p^{th}, q^{th} *and* r^{th} terms of both an A.P. and a G.P. be respectively a, b and c, then $\left(\frac{b}{c}\right)^a\left(\frac{c}{a}\right)^b\left(\frac{a}{b}\right)^c$ is

SECTION-III
(MATCHING LIST TYPE)

This section contains 4 questions, each having two matching lists (List-1 & List-II). The options for the correct match are provided as (A), (B),(C) and (D) out of which ONLY ONE is correct. **Marking scheme: +3 for correct answer, 0 if not attempted and -1 in all other cases.**

Section 3 (Max Marks: 12)

- This Section contains two (02) List-Match sets.
- Each List-Match set has TWO (02) Multiple-choice Questions
- Each List-Match set has two lists: List 1 and List 2
- List 1 has four entries (I), (II), (III), (IV) and List 2 has six entries (P), (Q), (R), (S), (T) and (U)
- Four options are given in each multiple choice question based on List 1 and List 2 and Only ONE of these four options satisfies the condition asked in the Multiple Choice Question.

Answer to each question will be evaluated according to the following marking scheme

Full Marks: +3 (If only the option corresponding to the correct combination is chosen))

Zero Marks: 0 (If none of the options is chosen (i.e. the question is unanswered)

Negative Marks: -1 (In all other cases)

Answer Q.51 and Q.52 by appropriately matching the list based on the information given in the paragraph

Let $f(x)=sin(\pi\, cosx)$ and $g(x)=cos(2\pi\, sinx)$ be two functions defined for $x>0$. Define the following sets whose elements are written in the increasing order:

$X=\{x:f(x)=0\}, Y=\{x:f'(x)=0\},\ Z=\{x:g(x)=0\}, W=\{x:g'(x)=0\}$

List – I contains the set X, Y, Z and W. List – II contains some information regarding these sets.

List – I		**List – II**
(I)	X	(P) $\supseteq\left\{\frac{\pi}{2},\frac{3\pi}{2},4\pi,7\pi\right\}$
(II)	Y	(Q) an arithmetic progression
(III)	Z	(R) NOT an arithmetic progression
(IV)	W	(S) $\supseteq\left\{\frac{\pi}{6},\frac{7\pi}{6},\frac{13\pi}{6}\right\}$
		(T) $\supseteq\left\{\frac{\pi}{3},\frac{2\pi}{3},\pi\right\}$
		(U) $\supseteq\left\{\frac{\pi}{6},\frac{3\pi}{4}\right\}$

51 Which of the following is the only CORRECT combination?

A. (II), (R), (S) **B.** (I), (Q), (U) **C.** (II), (Q), (T) **D.** (I), (P), (R)

52 Which of the following is the only CORRECT combination?

A. (IV), (P), (R), (S) **B.** (III), (P), (Q), (U)

C. (IV), (Q), (T) **D.** (III), (R), (U)

Answer Q.53 and Q.54 by appropriately matching the list based on the information given in the paragraph

Let the circles $C_1: x^2+y^2=9$ and $C_2:(x-3)^2+(y-4)^2=16,$ intersect at the points X and Y.

Suppose that another circle $C_3:(x-h)^2+(y-k)^2=r^2$ satisfies and following conditions:

(i) centre of C_3 is collinear with the centres of C_1 and C_2

(ii) C_1 and C_2 both lie inside C_3, and

(iii) C_3 touches C_1 at M and C_2 at N.

Let the line through X and Y intersect C_3 at Z and W, and let a common tangent of C_1 and (EQTN) be a tangent to the parabola $x^2=8\alpha y$.

There are some expressions given in the List–I whose values are given in List–II below:

List – I	**List – II**
(I) $2h+k$	(P) 6
(II) $\dfrac{\text{Length of } ZW}{\text{Length of } XY}$	(Q) $\sqrt{6}$
(III) $\dfrac{\text{Area of triangle } MZN}{\text{Area of triangle } ZMW}$	(R) $\dfrac{5}{4}$
(IV) α	(S) $\dfrac{21}{5}$
	(T) $2\sqrt{6}$
	(U) $\dfrac{10}{3}$

53 Which of the following is the only CORRECT combination?

A. (II), (T) **B.** (II), (Q) **C.** (I), (U) **D.** (I), (S)

54 Which of the following is the only INCORRECT combination?

A. (IV), (S) **B.** (III), (R) **C.** (IV), (U) **D.** (I), (P)